

Lab 1: Login and data consumption

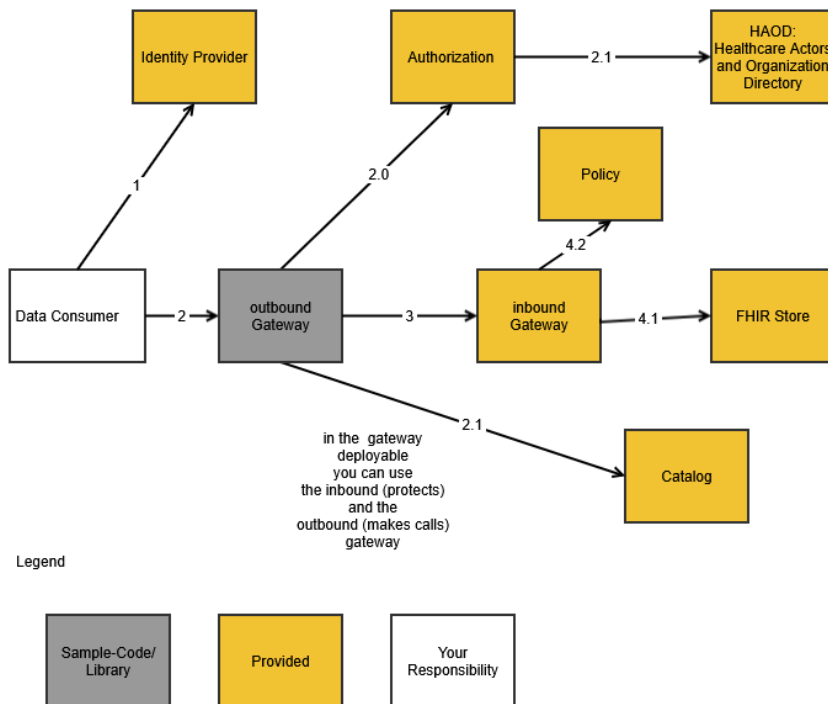
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Context

We are working in the context of SwissHDS. The SwissHDS reference architecture describes the conceptual model. Primary system vendors want to allow healthcare professionals to be logged into both their system and SwissHDS (or use a same login). These logins into both systems allow such healthcare professionals to access data from the SwissHDS in a seamless way.

We have set up a simplified SwissHDS reference environment. **It is not designed for real person data — only for test data. Also credentials such as tokens used must not be able to access production data. The interfaces and UIs are simplified.**

The following picture illustrates the situation:



Goals

- Healthcare professionals have a token to access the SwissHDS readily available when working in the primary system.
- The login process is seamless and not intrusive
- Healthcare professionals can get data from SwissHDS data stores and add it to their primary system

Steps

1) Get an AuthN Login Token for the healthcare professional & keep it current in your primary system

Select a strong IdP (as defined in the reference architecture) that is representative for typical customers and supports our IdP requirements.

You can use our simple demo IdP (a Keycloak instance) in case this helps your integration. Refer to [Environment-Details for Labs](#) and [Keycloak Usage](#)

In an ideal SwissHDS world, healthcare professionals can get a strong AuthN Token while they are using the primary system without an additional step. Some system might associate a strong IdP with the regular primary system login (so there is no second login needed); or the login of the primary system is strong enough to qualify for SwissHDS.

For others we expect that we need a second login with a strong IdP. We expect that the token for such a secondary logins should be held alive for e.g. 4h to reduce the impact on users.

The protocol to integrate the IdP into your primary system is the Authentication Request of OpenID Connect. We show an example login redirect in [Keycloak Usage](#).

We name this first token AuthN Token (for authentication).

2) Access data products located in other data stores in the SwissHDS

With the AuthN token, you can access data of other data stores. For this, either use the functionality of the HDS Gateway (outbound part) or implement this same functionality in your primary system.

We describe here how to access data of the existing user "Jon Doe" (but feel free to access other data).

Needed prerequisites:

- Authorization Service: The IdP of the AuthN Token is registered in the Authorization Service (this is already done for Keycloak) If you want us to add your own IdP, you cannot do this alone (it needs new environment variables). Refer to the [Authorization Service Usage#InternalConfiguration](#) for the details of what config we need for your own IdP.
- HAOD Service: There needs to be an entry for all your test healthcare professionals in the HAOD. It provides details of the healthcare professionals for the system. Again: you can test with our users in keycloak or add your own. Refer to [HAOD Usage](#) for details on adding new users.
- Policy Service: Set an "active relationship" in the policy service for the healthcare professional and patient Jon Doe. Jon Doe or a healthcare professional in an active relationship with a patient can see the patient's data. Refer to [Policy Service Usage](#)

To do this with the outbound gateway, you can use the operation GET /api/outbound/scan/fhir/{ResourceType}. Refer to [SwissHDS Gateway Usage](#) Via this convenience operation, you can invoke FHIR queries as you would on a FHIR store (but with authorization checks enabled).

3) Optional: Integrate the search/ access of step 2) in the UI and add it to the primary system

The data product search UI of could be integrated in the UI of the primary system. Relevant results of the search could be added as data to the primary system.

How elements are linked to each other

- AuthN token IdP Id: is linked via the HAOD person's IdP Id to a person (with a GLN Id or a HDS Id)
 - Patient John Doe

◦ AuthN Token IdP Id:

◦ HAOD-Entry:

550e8400-e29b-41d4-a716-446655440002 John Doe Patient (workaround) 761123610411353123

◦ AuthZ Token HDS Id:

- The document bundles's Patient Resource (resource.identifier.0.value) links to the patient's HDS Id.

○

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[+ New](#) [↻ Reload](#)

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- If you click on one of the patients, you see the data products of the patient:
Again, here are some yellow warnings of resources that are not visible.
You can click on data product instances (the dark blue words) to see them.

Suche

Zugriff verweigert:

- **DENIED_BY_DENY_RULE** https://swisshds.d.c.bag.admin.ch/patient-arhiv/Bunder/758/_history/ - denied by deny rule 14 (/Bundle(searchset)[0]/Bundle(document)[0]/Composition/subject)
- **DENIED_BY_DENY_RULE** https://swisshds.d.c.bag.admin.ch/patient-arhiv/Bunder/757/_history/ - denied by deny rule 13 (/Bundle(searchset)[5]/Bundle(document)[0]/Composition/subject)

Jane Doe - 05.05.1995 letzte 12 Monate letzten 5 Jahre von - bis

Datenprodukte (22/22) ~

ZEITRAUM	DATENLIEFERANT	GESUNDHEITSDATEN ☐ 2020 ORGANISATION/PRACITIONNER
07.03.2026 - 08.03.2026	Spital B	Patientenzusammenfassung (Patient Summary) Patientenzusammenfassung (EPR) (Patient Summary (EPR)) Patientenzusammenfassung (EPR) (Patient Summary (EPR)) Patientenzusammenfassung (EPR) (Patient Summary (EPR)) Patientenzusammenfassung (EPR) (Patient Summary (EPR)) Patientenzusammenfassung (EPR) (Patient Summary (EPR))
05.03.2026 - 06.03.2026	Spital A	Patientenzusammenfassung (Patient Summary) Patientenzusammenfassung (EPR) (Patient Summary (EPR))
21.02.2026	Spital A	Medikamentenabgabe
20.02.2026	Spital B	Laborbericht Zuweisungsschreiben (Orthopädischer Kontrollbericht und Rückzuweisung)
09.01.2026	Spital A	Rezept
15.12.2025 - 17.12.2025	Spital A	Laborbericht Laborbericht Laborbericht Medikamentenliste Medikationskarte (Medikationsplan)

- Use the developer console and tools of the browser
- Clicking on the top-right name of the currently logged-in user shows you the tokens (AuthN and AuthZ tokens)
- The policy service allows to test access rights
- In case you are proficient in Java, you could debug the gateway
- We try to list all the steps to access a Data Product Instance. You could go through this checklist in case of problems:
 1. The Login with your IdP works & an AuthN token of a practitioner, admin or patient is correctly established and returned
 2. The Authorization Service correctly translates the AuthN token to an AuthZ token, with all the expected information (from the HAOD)
 3. You look for one or multiple type of Data Products. Does the Data Product Catalog return all the URLs of the expected stores?
 4. Is the provider identity = URL of your inbound gateway configured both a) in your inbound gateway and b) in the System tab of the policy service? <https://swisshds.u.c.bag.admin.ch/policy/system>
 5. Are the FHIR query requests sent correctly to the URL of each inbound gateway?
 6. The outbound gateway makes the request on the underlying data store
 7. The inbound gateway validates that the holder of the AuthZ has the right to access the Data Product Instance. For this, it queries the Policy service.
 - a. You can yourself test the Policy service in its "Try Patient Access Evaluation" section <https://swisshds.u.c.bag.admin.ch/policy/patient>
- Error messages that some data stores were not found: If another team adds its own data store in the catalog, it is possible that your data consumer or the outbound gateway does not have network visibility of that store. This is expected in this lab setting. It is also expected in the future SwissHDS (with many stores, some stores will always be unavailable). So you should not be bothered by such messages. The outbound gateway handles this in a graceful way: It indicates what stores were not available (with a 403 error code just for this part of the result) but it still returns the results found (with a top-level return code that is OK).
- The outbound gateway currently has a limitation: it also returns its own data products.

Infrastructure and Support

Key documents are located in the GitHub project <https://github.com/SwissHDS/swiss-hds-labs> and are sent to you by email. If anything is unclear, Issues can be created in the same Github repository.

Questions

1. How much effort did it take you to finish the lab?
2. What IdP(s) did you use? Why?
3. What are your experiences/ Other remarks?
4. Screenshots/ Movie that shows your integration
5. Improvement ideas for SwissHDS?